

KnowSelf: Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

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Enabling agents to autonomously regulate knowledge utilization through situational self-awareness

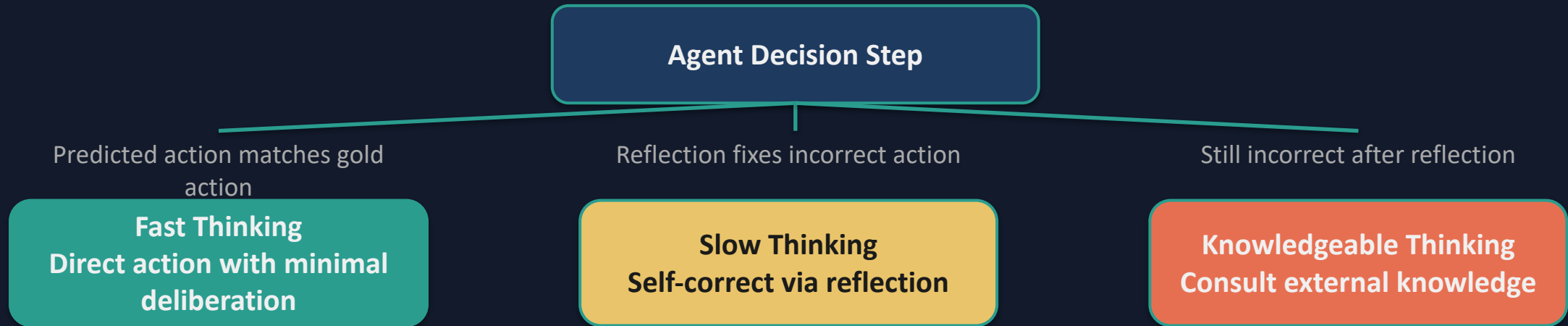
Current Agent Training Uses 'Flood Irrigation' — Indiscriminate Knowledge Injection Hurts Efficiency

- Direct trajectory imitation (e.g., ReAct) — blindly copies gold trajectories without situational need.
- Trial-and-error refinement (e.g., Reflexion) — injects external feedback regardless of whether the agent can self-correct.
- Knowledge-augmented planning (e.g., KnowAgent, WKM) — floods every step with domain knowledge, even when unnecessary.

Result: fragile planning, pattern collapse on unexpected inputs, and unnecessarily high inference costs.

Humans possess situational self-awareness — the metacognitive ability to assess whether they can handle a situation directly, need to pause and reflect, or must seek external help. KnowSelf brings this capability to LLM agents.

Three Situational Modes: Fast Thinking, Slow Thinking, and Knowledgeable Thinking



At each step, a heuristic criterion classifies the situation and inserts a special token.

- Fast Thinking → minimal compute, no external calls.
- Slow Thinking → reflection token triggers internal deliberation.
- Knowledgeable Thinking → knowledge token triggers retrieval from an external knowledge base.

This taxonomy mirrors human metacognition: act when confident, reflect when uncertain, seek help when stuck.

KnowSelf Framework: Data Construction → Two-Stage Training → Self-Aware Inference

Step 1: Self-Awareness Data Construction

Agent self-explores environments. Heuristic criterion classifies each step into fast / slow / knowledgeable. Special tokens are inserted to mark the situation type.

Step 2: Two-Stage Training

Supervised Fine-Tuning (SFT) teaches initial self-awareness patterns. Reward-based Policy Optimization (RPO) further refines generation of appropriate special tokens.

Step 3: Self-Aware Inference

During deployment, the agent generates special tokens that trigger direct action, reflection, or knowledge retrieval from an external knowledge base.

Step 1

Self-awareness Data Construction

Task: put a clean egg in microwave

Action: go to fridge 1
Observation: The fridge 1 is closed.
Action: open fridge 1
Observation: The fridge 1 is open. In it, you see a cup 3, a cup 1, a lettuce 1, an egg 2.

Action: close fridge 1

Action: take egg 2 from fridge 1

Reflection <> I need a clean egg, but egg 2 is not clean. I need to search for another place to find a clean egg.</>
Action: go to drawer 1

Reflection <> Oh, it seems that I missed the egg. I need to pick it up and clean it now.</>
Action: take egg 2 from fridge 1

Step 2

Self-awareness Learning



Step 3

Self-awareness Inference

Task: put two newspapers in drawer

Action: go to sofa 1
Observation: On the sofa 1, you see a creditcard 2, a newspaper 1.
Action: take newspaper 1 from sofa 1
Observation: You pick up the newspaper 1 from the sofa 1.
Action: go to counter 2
Observation: On the counter 2, you see a newspaper 2.

Fast Thinking
Action: take newspaper 2 from counter 2

Reflection <> Wait, let's check the action. Sorry, I made a mistake. I need put the first newspaper before

Knowledgeable Thinking
Knowledge <k> When the agent has taken a target object and needs to take additional items to fulfill a task, the agent should first place the current object at its target
selection

KnowSelf Correctly Heats a Mug and Places a Saltshaker Where Baselines Fail

Task: put a hot mug in cabinet

OpenAI-O1 incorrectly tries to take an egg from the microwave after opening it.

KnowSelf (Llama-8B) emits a Reflection token, realizes it should heat the mug using the open microwave, and succeeds.

Task: put a saltshaker in drawer

DeepSeek-R1 reflects that the drawer is open and tries to close it first — which fails.

KnowSelf applies a Knowledge token, noting that the agent should place the object without removing unrelated items, and succeeds.

(a) Task: put a hot mug in cabinet

.....
Action: take mug 1 from countertop 1
Observation: You pick up the mug 1 from the countertop 1.
Action: go to microwave 1
Observation: The microwave 1 is closed.
Action: open microwave 1
Observation: You open the microwave 1. In it, you see an egg 1.
[Knowledge]<k>If the agent is in a situation where it needs to modify the temperature of an object using a receptacle (e.g., a fridge for cooling), it should directly perform the specific action like 'cool' or 'heat' instead of

OpenAI-O1

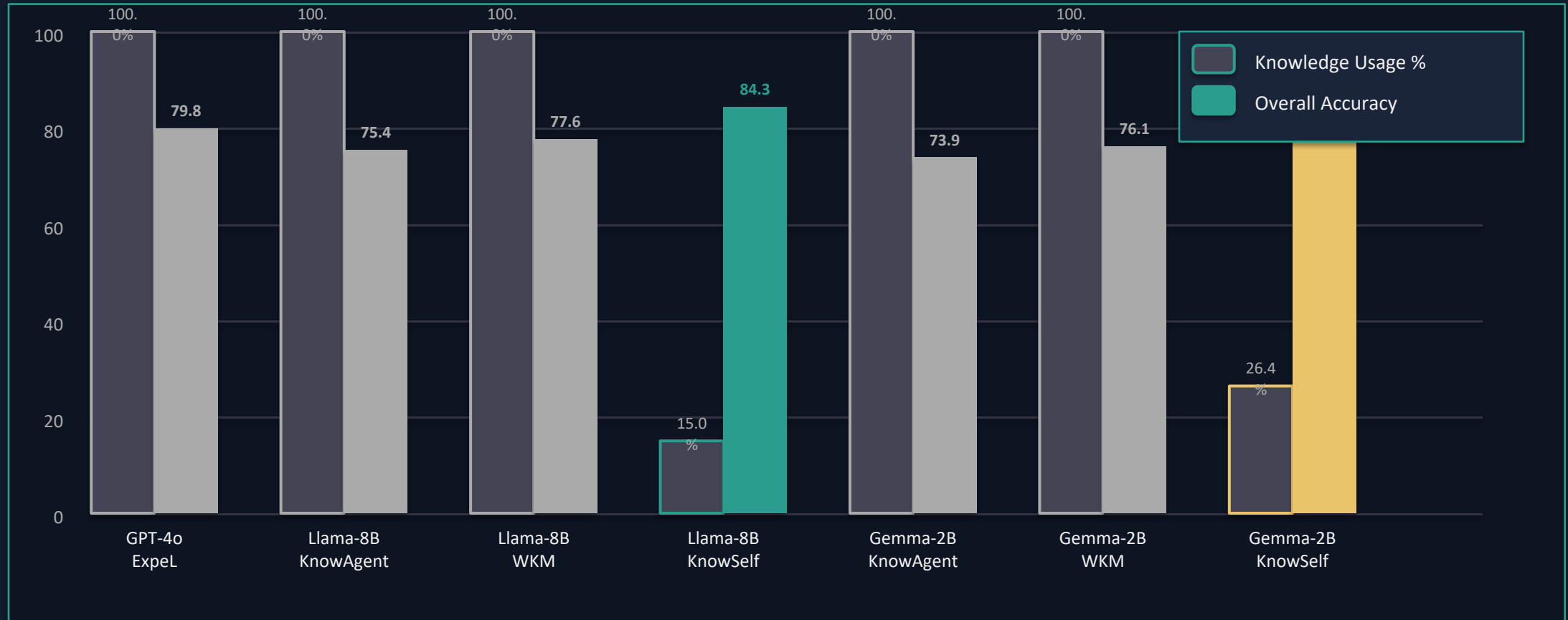
.....
Action: take mug 1 from countertop 1
Observation: You pick up the mug 1 from the countertop 1.
Action: go to microwave 1
Observation: The microwave 1 is closed.
Action: open microwave 1
Observation: You open the microwave 1. In it, you see an egg 1.
Action: put mug 1 in/on microwave 1
[Reflection]<r>Wait, let's check the action. Let's think step by step. With the microwave open, I can heat the mug using the microwave.</r>

KnowSelf
(Llama-8B)

KnowSelf Achieves 84.33% on ALFWorld with Llama-8B Using Only 15% Knowledge

Backbone	Method	Know %	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01 %	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41 %	87.50	93.55	73.91	76.19	83.33	52.94	79.85

Minimal Knowledge, Maximum Impact: KnowSelf's Efficiency Advantage Over Full-Knowledge Methods



- KnowSelf uses only 15% (Llama-8B) and 26% (Gemma-2B) knowledge queries vs. 100% for ExpeL / KnowAgent / WKM.
- Despite far fewer knowledge calls, KnowSelf achieves the highest overall accuracy on each backbone.
- Scaling: Gemma-2B KnowSelf (79.85%) matches GPT-4o ExpeL (79.85%), showing the approach scales down effectively.

Key Takeaways: Self-Awareness Is the Missing Piece in Agent Planning

1. Indiscriminate knowledge injection is wasteful and suboptimal. Methods flooding 100% of steps with knowledge often underperform selective use.
2. Agents can learn situational self-awareness via lightweight special-token mechanisms, requiring only heuristic labeling on self-explored trajectories.
3. KnowSelf with Llama-8B achieves 84.33% on ALFWorld, surpassing GPT-4o ReAct (76.12%) and GPT-4o ExpeL (79.85%).
4. Only 15–26% knowledge usage is needed versus 100% for competing methods, dramatically reducing inference cost without sacrificing accuracy.
5. The paradigm generalizes across model scales — even Gemma-2B reaches 79.85%, matching GPT-4o ExpeL and exceeding all Gemma-2B baselines.

Future Directions: Extend the three-situation taxonomy to multi-agent coordination, continuous online adaptation, and broader embodied AI benchmarks.